FPGA Synthesis and Prototyping Lab: 42
Frequency Dividers

The purpose of this experiment is to Synthesis and Check the behavior of the Frequency Dividers mentioned in Lab 25, 26.

LAB25:
The purpose of this experiment is to introduce the design of Divide by 2 Circuit
D-type Flip-Flop is as a binary divider, for Frequency Division or as a "divide-by-2" counter. Here the inverted output terminal Q (NOT-Q) is connected directly back to the Data input terminal D giving the device "feedback" as shown below.

CODE:
```
module feqdiv2(
input rst ,
output feqhalf , q_bar );
 wire clk ;
  SB_HFOSC osc_int(
      .CLKHFPU(1'b1),
     .CLKHFEN(1'b1),
      .CLKHF(clk)
   );
  defparam osc_int.CLKHF_DIV = "0b10"; //12MHZ
  reg d = 0 ;
  always @(posedge clk ) begin
  if (rst ) begin
  feqhalf <= 0 ;
  end
  else begin
      feqhalf <= d ;
        d <= q_bar ;
        q_bar <= ~feqhalf ;
        end
        end
        endmodule
```

PCF

set_io rst 37

set_io q_bar 36

set_io feqhalf 35

COMMANDS TO SYTH WITH ICE40

```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_42$ yosys -p "synth_ice40 -top feqdiv2 -json feqdiv2.json" feqdiv2.v

 /----------------------------------------------------------------------------\
 |                                                                            |
 |  yosys -- Yosys Open SYnthesis Suite                                       |
 |                                                                            |
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 |                                                                            |
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 Yosys 0.9 (git sha1 1979e0b)


-- Parsing `feqdiv2.v' using frontend `verilog' --

1. Executing Verilog-2005 frontend: feqdiv2.v
Parsing Verilog input from `feqdiv2.v' to AST representation.
Generating RTLIL representation for module `\feqdiv2'.
```

SYNTH RESULTS

```
Removed 0 unused modules.

2.44. Printing statistics.

=== feqdiv2 ===

   Number of wires:                  7
   Number of wire bits:              7
   Number of public wires:           5
   Number of public wire bits:       5
   Number of memories:               0
   Number of memory bits:            0
   Number of processes:              0
   Number of cells:                  6
     SB_DFFE                         2
     SB_DFFSR                        1
     SB_HFOSC                        1
     SB_LUT4                         2

2.45. Executing CHECK pass (checking for obvious problems).
checking module feqdiv2..
found and reported 0 problems.

2.46. Executing JSON backend.

Warnings: 3 unique messages, 3 total
End of script. Logfile hash: 69a0acbc2b
CPU: user 0.28s system 0.03s, MEM: 45.66 MB total, 40.23 MB resident
Yosys 0.9 (git sha1 1979e0b)
Time spent: 51% 10x read_verilog (0 sec), 17% 1x share (0 sec), ...
```

IMPLIMENTATIONS

```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_42$ nextpnr-ice40 --up5k --package sg48 --json feqdiv2.json --pcf cons.pcf --asc feqd
iv2.asc
Info: constrained 'rst' to bel 'X13/Y31/io0'
Info: constrained 'q_bar' to bel 'X9/Y31/io1'
Info: constrained 'feqhalf' to bel 'X12/Y31/io1'
```

BIN FILE PROG:

```
Info: Program finished normally.
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_42$ icepack feqdiv2.asc feqdiv2.bin
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_42$ iceprog feqdiv2.bin
init..
cdone: high
reset..
cdone: low
flash ID: 0xEF 0x40 0x16 0x00
file size: 104090
erase 64kB sector at 0x000000..
erase 64kB sector at 0x010000..
programming..
done.
reading..
VERIFY OK
cdone: high
Bye.
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_42$
```

OUTPUTS